Detonation and Decay:

The Physics and Persistent Hazards of Operation Redwing at Bikini and Enewetak Atolls

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Introduction

Throughout the spring and summer of 1956, the United States executed a test series, exploding seventeen nuclear devices at the Atomic Energy Commission's Pacific Proving Ground at Enewetak and Bikini Atolls.[1] Named *Operation Redwing*, the 1956 test series was the sixth of the 67 effectuated in the Marshall Islands during the onset of the Cold War.[2] Though the principle aim of Operation Redwing was to evaluate second-generation thermonuclear weapons, fission devices (both as primaries for hydrogen bombs and as small tactical air-defense armaments) were also tested.[3] Of the 1956 *Redwing* tests, the second test, *Cherokee*, marked the first U.S. airdrop of a deployable hydrogen bomb on May 21, 1956.[4]

Beyond proving delivery methods, *Operation Redwing* confronted another imperative: forecasting each detonation's total energy release (its "yield") and limiting the resulting radioactive fallout.[5] *Operation Castle*'s (1954) unexpectedly high yields prompted *Operation Redwing* (1956) to implement an 'energy budget,' strictly capping total and fission-derived yields.[6] Fission, primarily the 'fast' fission of a thermonuclear device's natural uranium tamper, boosts yield and constitutes the majority of its fallout, unlike the comparatively clean fusion process.[7] Although the bomb's mechanical and physical properties are relevant to understanding *how* radiative fallout was produced on Bikini and Enewetak Atolls, this paper concentrates on

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[1]United States, Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests: Nuclear Test Personnel Review*, prepared by the Defense Nuclear Agency as Executive Agency for the Department of Defense (Washington, DC: U.S. Government Publishing Office, 1956), PDF, https://www.govinfo.gov/content/pkg/GOVPUB-D15-PURL-gpo222811/pdf/GOVPUB-D15-PURL-gpo222811.pdf, 21.
[2]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 21.
[3]"Operation Redwing," Wikipedia, last edited May 20, 2024, accessed May 15, 2025, https://en.wikipedia.org/wiki/Operation_Redwing.
[4]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 22.
[5]"Operation Redwing," Wikipedia.
[6]"Operation Redwing," Wikipedia.
[7]"Operation Redwing," Wikipedia.

the physics of radiation more broadly and will briefly cover the physics by which that fallout wreaks destruction on biological matter.

Radiation Physics

Atomic Structure

As defined by the *United States Department of Energy*, "radioactivity is the release of energy from the decay of nuclei of certain kinds of atoms and isotopes."[8]

The atom consists of three fundamental subatomic particles: protons, neutrons, and electrons.[9] However, as radioactivity is a phenomenon of the nucleus, the electron will not be explained. The atomic nucleus, located at the center of an atom, is a tiny bundle of bound neutrons and protons which are referred to collectively as nucleons.[10]

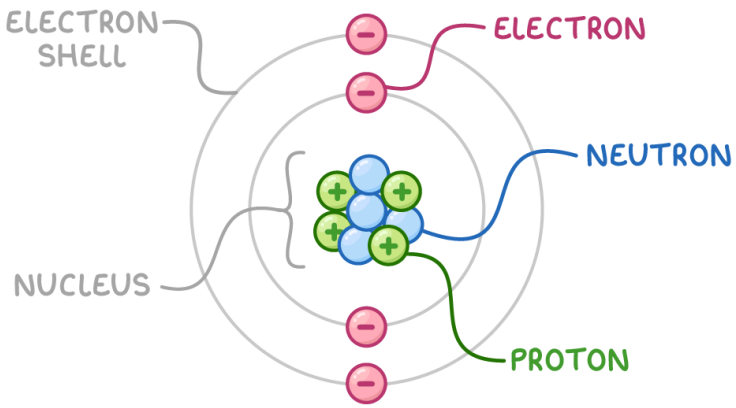

Figure 1. Diagram of the atom's structure.[11]

Radioactive Nuclei

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[8]U.S. Department of Energy, "DOE Explains…Radioactivity," DOE Office of Science, accessed May 18, 2025, https://www.energy.gov/science/doe-explainsradioactivity.
[9]Cognito Education, "A-Level Chemistry CIE," Cognito Education, accessed May 18, 2025, https://cognitoedu.org/coursesubtopic/c3-alevel-cie_cCQpEOkI.
[10]DOE, "DOE Explains…Radioactivity."; Douglas C. Giancoli, *Physics: Principles with Applications*, 5th ed. (Upper Saddle River, N.J.: Prentice Hall, 1998), chap. 30, 858.
[11]Cognito Education, "A-Level Chemistry CIE.

In radioactive nuclei, the nucleus is not stable: the nuclear force between nucleons, protons and neutrons is weak, causing the atomic nucleus to fail to stay together. As protons are positively charged, they experience electrostatic repulsion from one another. The opposing force to the electrostatic force is called a nuclear force.[12] Since stable nuclei *do* stay together, the nuclear force acts as an attractive force between all nucleons, protons, and neutrons alike with greater magnitude than the electrostatic force by which the positively charged protons are repelled from one another.[13] If the nuclear force is not stronger than the electric force, the nucleus will *not* stay together.[14] This weak nuclear force is a feature of radioactive nuclei.

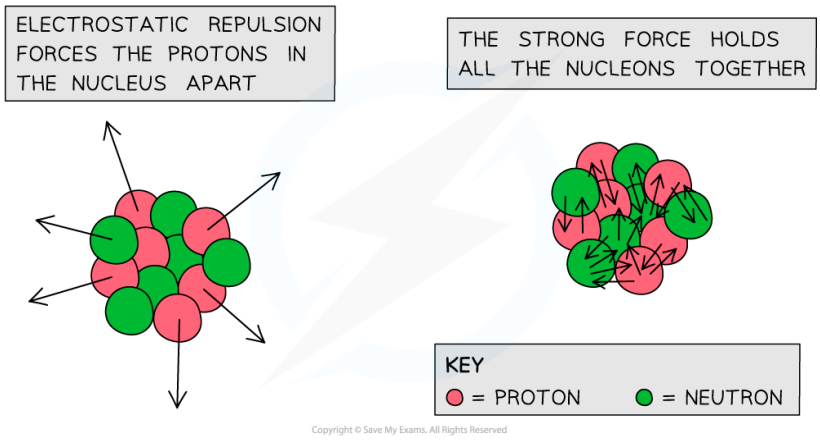

Figure 2. Comparison of strong and weak nuclear forces.[15]

Radioactive Decay

The instability arising when the nucleus cannot hold itself together manifests as radioactive decay, during which the nucleus emits particles or energy to achieve a more stable state. This emission of energy released due to the disintegration of an unstable nucleus is

[12]Giancoli, Physics: Principles with Applications, 862.
[13]Giancoli, Physics: Principles with Applications, 862.
[14]Giancoli, Physics: Principles with Applications, 862.
[15]Hanlin International Education, "AQA A Level Physics复习笔记2.1.3 Strong Nuclear Force," Hanlin International Education, July 7, 2022, accessed May 18, 2025, https://www.hanlin.com/archives/634677.

radiation.[16] Of the products emitted are alpha (α), beta (β), and gamma (γ) radiation, listed from least to most energetic.[17] Alpha radiation, being the least energetic, has inconsequential penetrating power; as Ernest Rutherford's late-1890s studies of the rays emitted in radioactivity revealed, alpha radiation could barely penetrate a piece of paper.[18] By comparison, beta radiation can traverse up to three millimeters of aluminum, while gamma radiation can pass through several centimeters of lead and still be detected on the other side, rendering it the most penetrating and energetic (and therefore the most harmful) form.[19]

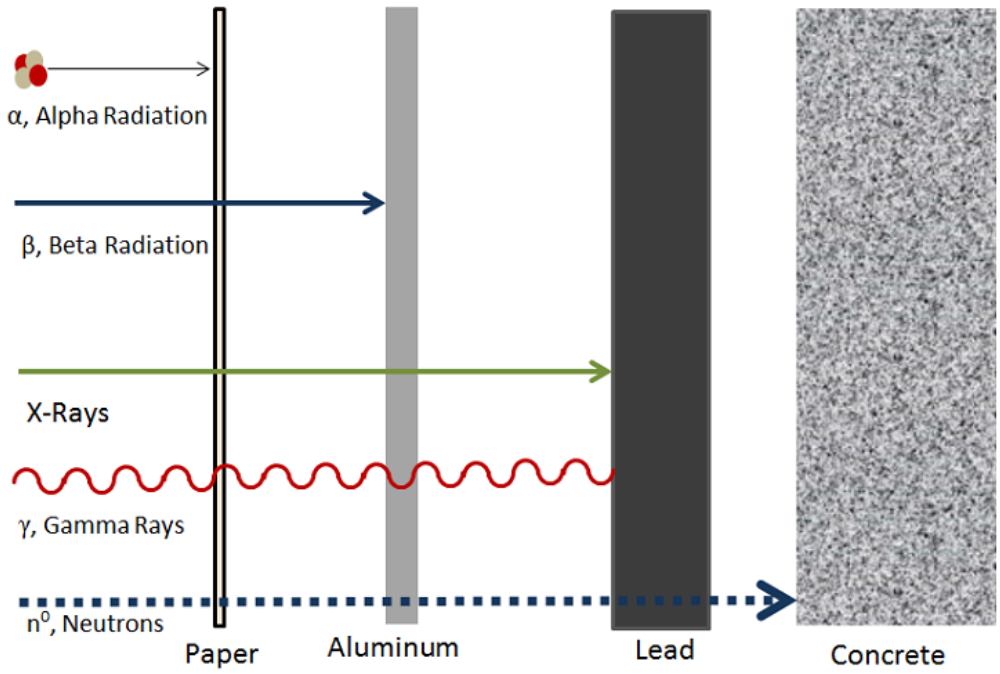

Figure 3. Different penetration levels of different products of decay.[20]

Gamma Radiation

[16]Giancoli, Physics: Principles with Applications, 863.
[17]Giancoli, Physics: Principles with Applications, 863.
[18]Giancoli, Physics: Principles with Applications, 863.
[19]Giancoli, Physics: Principles with Applications, 863.
[20]"Gamma Decay," *Energy Education*, accessed May 20, 2025, https://energyeducation.ca/encyclopedia/Gamma_decay#:~:text=Gamma%20decay%20is%20one%20type.

Not only does gamma radiation differ from alpha and beta radiation in that its penetrating power is much greater, the mechanics of the decay itself is distinct.[21] The mechanical differences are two-fold: rather than ejecting charged particles or altering an atom's structure or composition, gamma decay emits a high-energy photon—carrying neither mass nor charge and changing only the nucleus's energy state.[22] These properties underpin its extraordinary penetration, making the gamma component of the residual radiation experienced following the 1956 denotations a primary concern for *Operation Redwing.* [23]

Radiation Physics Applied

Nuclear Tests and Radiation Exposures

Prior to 1963, the denotation of nuclear devices was unconfined.[24] Thus, personnel could be exposed to initial radiation (the radiation emitted at the time of the explosion or for one minute thereafter) or residual radiation.[25] Considering that initial radiation is part of the violent nuclear explosion itself and that any observer exposed to it would be swept away by the lethal blast and thermal effects of the detonation, it is unlikely that an operation would intentionally place an observer in that area, rendering the potential exposure of residual radiation as more of the operations' concern.[26]

However, the neutron component of initial radiation contributed indirectly to the potential exposure of personnel.[27] Since neutrons are emitted in large numbers by nuclear weapon explosions, they have the property of altering non-radioactive materials (such as sodium, silicon,

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[21]"Gamma Decay," *Energy Education*.
[22]"Gamma Decay," *Energy Education*.
[23]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 141.
[24]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[25]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[26]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[27]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.

manganese, and iron) into radioactive ones.[28] Named 'activation,' this process multiplies the quantity of radioactive products produced as residual radiation.[29]

In addition to the residual radiation produced by activation, the lighter atoms released in the nuclear explosion process contribute to the overall residual radiation.[30] When the fissioning atoms of heavy elements (uranium and plutonium) split into lighter atoms, energy release ensues: the lighter atoms are radioactive themselves, decaying into successive generations of new isotopes.[31] This decay happens very rapidly in the moments after detonation, then slows over time, though trace levels of radioactivity persist indefinitely, with myriad ramifications.[32]

Even decades later—Bikini Atoll still exceeded the 100 mrem/year safety standard in 2016, and parts of Enewetak remained uninhabitable until 1980 after an extensive cleanup—these tests had devastating consequences: the nuclear radiation produced inflicted both acute and chronic harm on human health, laid waste to local ecosystems, and created political tensions between the United States and the Marshall Islands.[33]

Passage of Radiation Through Matter; Biological Damage

Radiation harms living organisms primarily by ionizing their cells. Each form of radiation achieves this by ejecting electrons from atoms.[34] Alpha and beta particles ionize atoms by exerting electric forces that pull or push electrons until they are dislodged, while gamma

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[28]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[29]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[30]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[31]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[32]Defense Nuclear Agency, *United States Atmospheric Nuclear Weapons Tests*, 24.
[33]Thomas Sumner, "Bikini Atoll Radiation Levels Remain Alarmingly High," *Science News*, June 6, 2016, https://www.sciencenews.org/article/bikini-atoll-radiation-levels-remain-alarmingly-high.; Children of Atomic Veterans, "Enewetak Cleanup Veterans," Children of Atomic Veterans, accessed May 21, 2025, https://childrenofatomicveterans.org/international-outreach/atomic-weapons-testing/enewetak-cleanup-veterans/.
[34]Giancoli, Physics: Principles with Applications, 899.

photons induce ionization through the photoelectric effect and Compton scattering.[35] When a gamma photon has more than 1.02 MeV of energy, it can undergo pair production, producing an electron and a positron that then initiate successive rounds of ionization.[36] If those ejected electrons are bonding electrons, the molecule may break apart or change shape, preventing its normal function or causing harmful effects.[37] Although the loss of a single protein molecule is typically harmless—cells have multiple copies and can produce replacements from the encoding gene—excessive radiation can destroy so many molecules that the cell's repair machinery is overwhelmed and the cell ultimately dies.[38] When enough cells are damaged, tissues or even whole organs can fail and put the organism at risk.[39] That level of injury may come from a single intense burst of radiation or from long-term, lower-level exposure that accumulates DNA damage and disrupts cellular function.[40] Because each cell carries only one copy of its DNA, any change to that DNA can disrupt a gene and prevent its encoded protein (or other product) from ever being produced.[41] Surviving cells may carry DNA damage and continue dividing, producing increasingly defective daughter cells that harm the organism.[42] In this way, radiation can trigger cancer, defined by the rapid, uncontrolled proliferation of abnormal cells.[43]

A study indexed by the U.S. National Library of Medicine, *Radiation Doses and Cancer Risks in the Marshall Islands Associated with Exposure to Radioactive Fallout from Bikini and Enewetak Nuclear Weapon Tests*, found that exposure to radioactive fallout, particularly in the

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[35]Giancoli, Physics: Principles with Applications, 899.
[36]Giancoli, Physics: Principles with Applications, 899.
[37]Giancoli, Physics: Principles with Applications, 899.
[38]Giancoli, Physics: Principles with Applications, 899.
[39]Giancoli, Physics: Principles with Applications, 899.
[40]Giancoli, Physics: Principles with Applications, 899.
[41]Giancoli, Physics: Principles with Applications, 899.
[42]Giancoli, Physics: Principles with Applications, 899.
[43]Giancoli, Physics: Principles with Applications, 899.

period immediately following a large detonation, led to high radiation doses and a substantial increase in cancer risk among the Marshallese population.[44]

Conclusion

Although Operation Redwing, along with the 67 other detonations that took place in the Marshall Islands, implemented cleanup programs, none of those efforts were effective in preventing exposure to nuclear radiation. Given these consequences, it would be illogical to conduct further nuclear weapons testing in the future.

[44]Steven L. Simon, André Bouville, Charles E. Land, and Harold L. Beck, "Radiation Doses and Cancer Risks in the Marshall Islands Associated with Exposure to Radioactive Fallout from Bikini and Enewetak Nuclear Weapons Tests: Summary," Health Physics 99, no. 2 (2010): 105, https://pmc.ncbi.nlm.nih.gov/articles/PMC4042840/.

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